Recitation 3.

Problem 1. Let $X_1, \ldots, X_n$ be independent $N(\theta, \theta)$ random variables with $\theta > 0$.

(1) Find the maximum likelihood estimator of θ. (Hint: It is the solution of a quadratic form.)

(2) Using the Law of Large Numbers, can you deduce that this mle is consistent?

(3) Find the asymptotic distribution of the MLE using the delta method and the asymptotic normality of (a properly standardized version of) $\sum_{i=1}^{n} X_i^2$.
You may use the fact that $E[X^4] = \theta^4 + 6\theta^3 + 3\theta^2$.

Problem 2. Let $X_1, \ldots, X_n$ be independent random variables with common density

$$f(x) = \begin{cases} \theta^{-2} x e^{-x/\theta} & \text{if } x > 0; \\ 0 & \text{otherwise} \end{cases}$$

with $\theta > 0$ unknown. This distribution has mean $E(X) = 2\theta$ and variance $Var(X) = 2\theta^2$.

(1) Compute the method of moments estimator of θ.

(2) Compute the maximum likelihood estimator $\widehat{\theta}$ of θ.

(3) Compute the mean squared error of $\widehat{\theta}$.

Problem 3.

The stochastic vector (X, Y) has a uniform distribution on the circle

$$\mathcal{C} = \left\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq \theta^2\right\}.$$

Thus the joint density of (X, Y) is given by

$$f(x, y) = 1/(\pi\theta^2) \text{ if } (x, y) \in \mathcal{C} \text{ and } f(x, y) = 0 \text{ otherwise.}$$

We take a random sample $(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$, and define

$$R_i = \sqrt{X_i^2 + Y_i^2}.$$

(1) Find the maximum likelihood estimator of θ.

(2) Show that $P(R_1 \leq x) = (x/\theta)^2$ for $0 < x < \theta$.

(3) Is the MLE unbiased?

Solutions

Problem 1. Let $X_1, \ldots, X_n$ be independent $N(\theta, \theta)$ random variables with $\theta > 0$.

(1) *Find the maximum likelihood estimator of θ.*

We observe that

$$\begin{aligned}
\mathrm{lik}\,(\theta) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\theta}} \exp\left(-\frac{(X_i-\theta)^2}{2\theta}\right) = (2\pi\theta)^{-n/2} \exp\left(-\sum_{i=1}^{n} \frac{(X_i-\theta)^2}{2\theta}\right);\\
\ell\,(\theta) &= -\frac{n}{2}\log 2\pi - \frac{n}{2}\log\theta - \frac{\sum_{i=1}^{n}(X_i-\theta)^2}{2\theta};\\
\ell'\,(\theta) &= -\frac{n}{2\theta} - \frac{1}{2}\sum_{i=1}^{n}\left(1-\theta^{-2}X_i^2\right).
\end{aligned}$$

Setting $\ell'\,(\theta)$ to 0 yields

$$\begin{aligned}
-\frac{n}{2\theta} - \frac{1}{2}\sum_{i=1}^{n}\left(1-\theta^{-2}X_i^2\right) = 0 &\iff n - \theta^{-2}\sum_{i=1}^{n} X_i = -n\theta^{-1}\\
&\iff \theta^2 + \theta - \frac{1}{n}\sum_{i=1}^{n} X_i^2 = 0\\
&\iff \theta = -\frac{1}{2} \pm \sqrt{\frac{1}{4} + \frac{\sum_{i=1}^{n} X_i^2}{n}}.
\end{aligned}$$

Since $\theta > 0$, it follows that

$$\widehat{\theta}_{MLE} = -\frac{1}{2} + \sqrt{\frac{1}{4} + \frac{\sum_{i=1}^{n} X_i^2}{n}}$$

by noting that the other solution is less than 0.

(2) *Using the Law of Large Numbers, can you deduce that the mle is consistent?*

Set $Y_i = X_i^2$, $i = 1, \ldots, n$. Since

$$\mathbb{E}[\overline{Y}_n] = \mathbb{E}\left[X_1^2\right] = \mathrm{Var}\left[X_1\right] + \mathbb{E}\left[X_1\right]^2 = \theta + \theta^2,$$

the LLN implies that

$$\overline{Y}_n = \frac{1}{n}\sum_{i=1}^{n} X_i^2 \to_p \theta + \theta^2.$$

Note that $\widehat{\theta}_{MLE} = g\left(\frac{1}{n}\sum_{i=1}^{n} X_i^2\right)$, for the continuous function

$$g(x) = -\frac{1}{2} + \sqrt{\frac{1}{4} + x}.$$

By the continuous mapping theorem, we have

$$\widehat{\theta}_{MLE} = g\left(\frac{1}{n}\sum_{i=1}^{n} X_i^2\right) \to_p g\left(\theta + \theta^2\right) = -\frac{1}{2} + \sqrt{\frac{1}{4} + \theta + \theta^2} = \theta.$$

Therefore, $\widehat{\theta}_{MLE}$ is consistent.

(3) *Find the asymptotic distribution of the MLE using the delta method and the asymptotic normality of (a properly standardized version of)* $\sum_{i=1}^{n} X_i^2$.

Use $\mathbb{E}[X_1^2] = \theta^2 + \theta$ and $\mathbb{E}[X_1^4] = \theta^4 + 6\theta^3 + 3\theta^2$ to obtain

$$\begin{aligned}\mathrm{Var}(Y_1) = \mathrm{Var}\left[X_1^2\right] &= \mathbb{E}\left[X_1^4\right] - \left(\mathbb{E}\left[X_1^2\right]\right)^2 \\ &= \theta^4 + 6\theta^3 + 3\theta^2 - \left(\theta^2 + \theta\right)^2 \\ &= 4\theta^3 + 2\theta^2.\end{aligned}$$

By the CLT,

$$\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} Y_i - \mathbb{E}[Y_1]\right) \to_d N\left(0, \mathrm{Var}[Y_1]\right)$$

whence

$$\sqrt{n}\left(\overline{Y}_n - \left(\theta + \theta^2\right)\right) \to_d N\left(0, 4\theta^3 + 2\theta^2\right).$$

Recall $\widehat{\theta} = g\left(\overline{Y}_n\right)$ and $g(\mathbb{E}[Y_1]) = \theta$, and note that

$$g'(x) = \frac{1}{\sqrt{4x+1}}$$

is continuous with

$$g'\left(\mathbb{E}[Y_1]\right) = g'\left(\theta+\theta^2\right) = \frac{1}{\sqrt{4\theta+4\theta^2+1}} \neq 0.$$

Hence, by the delta-method, the distribution of

$$\sqrt{n}(\widehat{\theta}_{MLE}-\theta) = \sqrt{n}\left(g\left(\overline{Y}_n\right) - g\left(\theta+\theta^2\right)\right)$$

converges to

$$N\left(0, \left[g'\left(\theta+\theta^2\right)\right]^2\left(4\theta^3+2\theta^2\right)\right) = N\left(0, \frac{2\theta^2}{1+2\theta}\right).$$

We conclude that the asymptotic variance of $\widehat{\theta}_{MLE}$ is

$$\frac{2\theta^2}{n\left(1+2\theta\right)}.$$

Problem 2. Let $X_1, \ldots, X_n$ be be independent random variables with common density

$$f(x) = \begin{cases} \theta^{-2}xe^{-x/\theta}, & \text{if } x>0; \\ 0, & \text{otherwise} \end{cases}$$

with with $\theta > 0$ unknown. Note that $\mathbb{E}(X) = 2\theta$ and $\mathrm{Var}(X) = 2\theta^2$.

(1) *Compute the method of moments estimator of θ.*

Since $\mu_1 = \mathbb{E}(X) = 2\theta$, we have $\theta = \mu_1/2$, whence $\widehat{\theta}_{MOM} = \overline{X}/2$.

(2) *Compute the maximum likelihood estimator $\widehat{\theta}$ of θ.*

We compute

$$\mathrm{lik}(\theta) \;=\; \theta^{-2n}\prod_{i=1}^{n} X_i \exp\left(-\frac{1}{\theta}\sum_{i=1}^{n} X_i\right)$$

$$\begin{aligned}\ell(\theta) &= -2n\log\theta + \sum_{i=1}^{n}\log X_i - \frac{1}{\theta}\sum_{i=1}^{n} X_i \\ \ell'(\theta) &= -\frac{2n}{\theta} + \frac{1}{\theta^2}\sum_{i=1}^{n} X_i.\end{aligned}$$

By solving $\ell'(\theta) = 0$, we find the MLE as

$$\widehat{\theta} = \overline{X}/2.$$

(3) *Compute the mean squared error of* $\widehat{\theta}$.

By definition,

$$\begin{aligned}\mathrm{MSE}(\widehat{\theta}) &= (\mathbb{E}[\widehat{\theta}] - \theta)^2 + \mathrm{Var}(\widehat{\theta}) \\ &= \left(\frac{\mathbb{E}(X_1)}{2} - \theta\right)^2 + \frac{1}{4n}\mathrm{Var}(X_1) \\ &= 0 + \frac{2\theta^2}{4n} = \frac{\theta^2}{2n}.\end{aligned}$$

Problem 3. (1) *Find the maximum likelihood estimator of* θ.

The likelihood is

$$\mathrm{lik}(\theta) = \frac{1}{\pi^n\theta^{2n}}\prod_{i=1}^{n}\mathbf{1}\{R_i^2 \le \theta^2\} = \frac{1}{\pi^n\theta^{2n}}\mathbf{1}\left\{\max_{1\le i\le n} R_i^2 \le \theta^2\right\}.$$

Since $1/\theta^{2n}$ is a decreasing function in θ, the likelihood is maximized at

$$\widehat{\theta} = \max(R_1, \ldots, R_n).$$

(2) *Show that* $\mathbb{P}(R_1 \le x) = (x/\theta)^2$ *for* $0 < x < \theta$.

For any $0 \leq r \leq \theta$, one has

$$\mathbb{P}\left(R_1 \leq r\right) = \mathbb{P}\left(X_1^2 + Y_1^2 \leq r^2\right) = \frac{1}{\pi\theta^2} \int_{x^2+y^2 \leq r^2} dxdy = \frac{\pi r^2}{\pi\theta^2} = \frac{r^2}{\theta^2}$$

as desired.

(3) *Is the MLE unbiased?*

We need to calculate $\mathbb{E}[\widehat{\theta}] = \mathbb{E}\left[\max(R_1, \ldots, R_n)\right]$. To this end, we note that for any $0 \leq r \leq \theta$,

$$\begin{aligned}
\mathbb{P}(\widehat{\theta} \leq r) &= \mathbb{P}\left(\max(R_1^2, \ldots, R_n^2) \leq r^2\right) \\
&= \mathbb{P}\left(R_1^2 \leq r^2, \ldots, R_n^2 \leq r^2\right) \\
&= \left[\mathbb{P}\left(R_1^2 \leq r^2\right)\right]^n && \text{(by the i.i.d. property)} \\
&= \frac{r^{2n}}{\theta^{2n}} && \text{(by part (2)).}
\end{aligned}$$

Hence, the density function of $\widehat{\theta}$ is

$$f(r) = 2n\theta^{-2n} r^{2n-1}, \quad \forall\ 0 \leq r \leq \theta$$

and conclude

$$\mathbb{E}\left[\widehat{\theta}\right] = \int_0^\theta 2nr^{2n}\theta^{-2n} dr = \frac{2n}{2n+1}\theta^{-2n} r^{2n+1}\Big|_0^\theta = \frac{2n}{2n+1}\theta$$

So $\widehat{\theta}$ is biased. This makes sense, as it always *under-estimates* θ. However, it is asymptotically unbiased.

Alternatively, since $\widehat{\theta}$ is non-negative, we don't have to compute the density, but instead use Problem 13 at page 167, to conclude

$$\mathbb{E}[\widehat{\theta}] = \int_0^\theta [1 - \mathbb{P}(\widehat{\theta} \leq r)] dr = \theta - \theta^{-2n} \frac{\theta^{2n+1}}{2n+1} = \frac{2n}{2n+1}\theta$$